

Build ARIMA or Prophet forecasting model

For this task, I have used a dataset related to **weather and temperature**. The main objective of this task was to perform **time series forecasting** using Python and libraries such as statsmodels and prophet. I have built **Prophet model**, used **Prophet library** for this.

As prophet library is not supported in jupyter notebook, I done this task in google colab.

First, I imported the required libraries like pandas, numpy, matplotlib, and prophet.

Then, I imported the dataset and performed basic checks like viewing the data, checking data types and understanding the structure. The date column was converted into datetime format and set as index.

After that, I selected **the temperature columns (maximum and minimum temperature)**. A line graph was plotted to learn the pattern of the data. The graph showed variations over time.

The dataset was divided into training and testing sets.

Prophet Model

A Prophet model was built using the dataset. The data was prepared by renaming columns as 'ds' (date) and 'y' (value).

The model was trained on the training data and predictions were made for the test data.

The forecast was plotted along with actual values. The Prophet model captured both trend and patterns more effectively.

Model performance was evaluated using:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)